

Practical no. 7

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Class :- CO3K – B

Subject :- OOP

Aim :- WAP to exchange values of two variable using friend function.

Program :-

```
#include<iostream.h>

#include<conio.h>

class B;

class A{

int x;

public:

void accept(){

cout<<"\nEnter the value for x :- ";

cin>>x;

}

friend void swap(A,B);

};

class B{

int y;

public:

void accept(){

cout<<"\nEnter the value for y :- ";

cin>>y;

}

friend void swap(A,B);

};

void swap(A a,B b){
```

```

cout<<"\nBefore swapping :- ";

cout<<"\nValue of x :- "<<a.x;

cout<<"\nValue of y :- "<<b.y;

int temp;

temp=a.x;

a.x=b.y;

b.y=temp;

cout<<"\nAfter swapping :- ";

cout<<"\nValue of x :- "<<a.x;

cout<<"\nValue of y :- "<<b.y;

}

void main(){

A a;

B b;

clrscr();

a.accept();

b.accept();

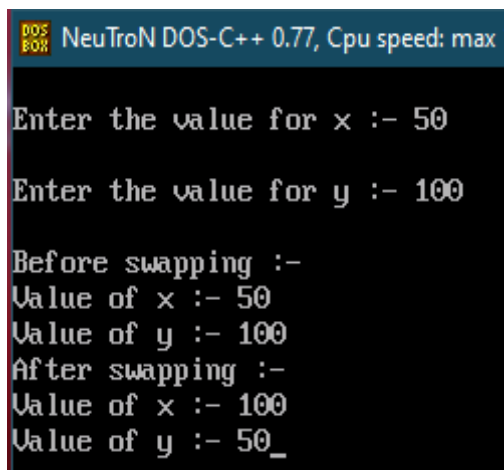
swap(a,b);

getch();

}

```

OUTPUT:-



The screenshot shows a DOS box window titled "NeuTroN DOS-C++ 0.77, Cpu speed: max". The program prompts the user to enter values for x and y. x is entered as 50 and y as 100. The program then displays the values before and after swapping. Before swapping, x is 50 and y is 100. After swapping, x is 100 and y is 50.

```

DOS
BOX NeuTroN DOS-C++ 0.77, Cpu speed: max

Enter the value for x :- 50

Enter the value for y :- 100

Before swapping :-
Value of x :- 50
Value of y :- 100
After swapping :-
Value of x :- 100
Value of y :- 50_

```

Aim :- WAP to create two classes test1 and test2 which stores marks of a student. Read value for class objects and calculate average of two test using friend function.

Program :-

```
#include<iostream.h>

#include<conio.h>

class Test2;

class Test1{

int m1;

public:

void accept(){

cout<<"\nEnter marks of 1st test :- ";

cin>>m1;

}

friend void avg(Test1,Test2);

};

class Test2{

int m2;

public:

void accept(){

cout<<"\nEnter marks of 2nd test :- ";

cin>>m2;

}

friend void avg(Test1,Test2);

};

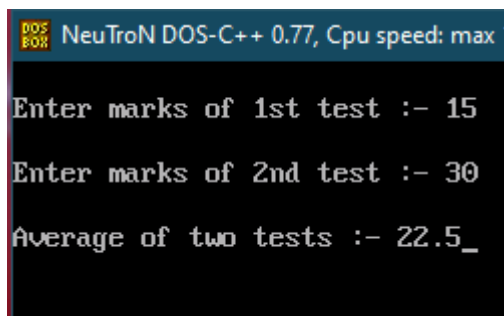
void avg(Test1 t1,Test2 t2){

float avg=(float)(t1.m1+t2.m2)/2;

cout<<"\nAverage of two tests :- "<<avg;
```

```
}  
  
void main(){  
  
Test1 t1;  
  
Test2 t2;  
  
clrscr();  
  
t1.accept();  
  
t2.accept();  
  
avg(t1,t2);  
  
getch();  
  
}
```

OUTPUT:-

A screenshot of a DOS-based C++ IDE window titled "NeuTroN DOS-C++ 0.77, Cpu speed: max". The window has a black background with white text. It shows the program's output: "Enter marks of 1st test :- 15", "Enter marks of 2nd test :- 30", and "Average of two tests :- 22.5_".

```
DOS FOR NeuTroN DOS-C++ 0.77, Cpu speed: max  
Enter marks of 1st test :- 15  
Enter marks of 2nd test :- 30  
Average of two tests :- 22.5_
```